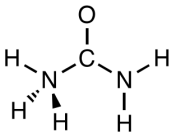
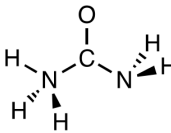
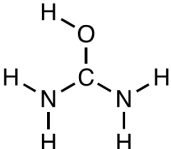
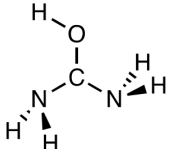
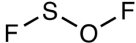
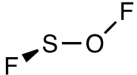
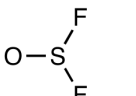
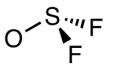
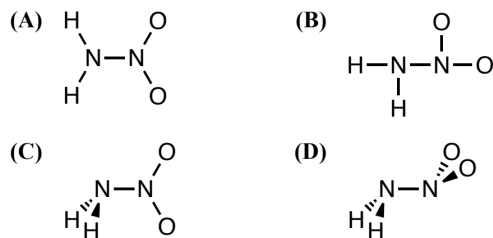


Bonding/Molecular Structure

- In which are the species arranged in increasing order of N-O bond length?
 (A) $\text{NO}^+ < \text{NO} < \text{NO}_3^-$ (B) $\text{NO}^+ < \text{NO}_3^- < \text{NO}$
 (C) $\text{NO} < \text{NO}^+ < \text{NO}_3^-$ (D) $\text{NO}_3^- < \text{NO}^+ < \text{NO}$
- Which statement does not accurately describe σ and π bonding?
 (A) σ bonds have no nodes containing the internuclear axis, while π bonds have one node containing the internuclear axis.
 (B) σ and π bonds are of similar strength in homodiatom molecules containing elements of the second and third period.
 (C) Triple bonds in heterodiatom molecules of the second and third period elements consist of one σ and two π bonds.
 (D) Formaldehyde, H_2CO , has three σ bonds and one π bond.
- What is the bond order in superoxide ion, O_2^- ?
 (A) 1.0 (B) 1.5 (C) 2.0 (D) 2.5
- Which is the best description of the geometry of XeOF_4 ?
 (A) Square pyramidal with the oxygen axial
 (B) Square pyramidal with the oxygen equatorial
 (C) Trigonal bipyramidal with the oxygen axial
 (D) Trigonal bipyramidal with the oxygen equatorial
- Which is the best description of the arrangement of the atoms in space in the protonated urea ion, $\text{H}_5\text{CN}_2\text{O}^+$?
 (A)  (B) 
 (C)  (D) 
- How many stereoisomers are there of the square planar complex $(\text{en})\text{PtClBr}$ ($\text{en} = 1,2\text{-ethylenediamine}$, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$)?
 (A) 1 (B) 2 (C) 3 (D) 4
- Which molecule is trigonal planar?
 (A) NH_3 (B) NF_3 (C) SO_3 (D) ClF_3
- Which molecule has the shortest N-O bond?
 (A) NO (B) NO_2 (C) N_2O (D) N_2O_4
- How many lone pairs are there in the Lewis structure of nitromethane, CH_3NO_2 ?
 (A) 3 (B) 4 (C) 5 (D) 7
- Which is the strongest acid in aqueous solution?
 (A) NH_4^+ (B) H_2O (C) PH_4^+ (D) H_2S
- Which is the best representation of the three-dimensional arrangement of the atoms in thionyl fluoride, SOF_2 ?
 (A)  (B) 
 (C)  (D) 
- Which ion is diamagnetic?
 (A) $\text{Fe}(\text{CN})_6^{4-}$ (B) CoCl_4^{2-}
 (C) $\text{Ni}(\text{H}_2\text{O})_6^{2+}$ (D) $\text{Cu}(\text{NH}_3)_4^{2+}$
- Which species has the shortest bond to nitrogen?
 (A) HCN (B) N_2O_4 (C) NO_2 (D) NO_2^-
- How many lone pairs are there in a valid Lewis structure of ozone, O_3 ?
 (A) 6 (B) 7 (C) 8 (D) 9
- Which species is linear?
 (A) H_2F^+ (B) FCO^+ (C) FNO (D) N_2F_2
- Anionic NO^- has a longer bond than neutral NO . Which is the best explanation for this difference?
 (A) NO^- has two unpaired electrons while NO has only one.
 (B) NO^- has a double bond while NO has a triple bond.
 (C) NO^- has one more σ antibonding electron than NO .
 (D) NO^- has one more π antibonding electron than NO .

Bonding/Molecular Structure

17. Which is the best representation of the three-dimensional arrangement of the atoms in nitramide, H_2NNO_2 ?



18. How many stereoisomers are possible for the square planar complex $\text{Pt}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)(\text{C}_2\text{O}_4)$?

(A) 1 (B) 2 (C) 3 (D) 4

19. Which species has the largest O–N–O angle?

(A) FNO_2 (B) NO_2 (C) NO_2^- (D) NO_3^-

20. Which molecule has the shortest bond?

(A) H_2 (B) N_2 (C) O_2 (D) F_2

21. Which molecules are described by molecular orbital theory as having a bond order of 2.0?

I. CO II. NO

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

22. Which is the best description of the geometry of IF_7 ?

(A) Heptagonal planar
(B) Hexagonal monopyramidal
(C) Pentagonal bipyramidal
(D) Capped trigonal prismatic

23. Carbon monoxide, CO , has a very small dipole moment. Which is the best explanation for this observation?

(A) The carbon-oxygen π bonds are polarized in the opposite direction from the carbon-oxygen σ bonds, so the bond dipoles tend to cancel each other out.
(B) Carbon and oxygen each have the same number of lone pairs despite the fact that carbon has fewer valence electrons than oxygen, partially canceling the bond dipoles.
(C) Neither carbon nor oxygen has a formal charge in the Lewis structure of CO , giving an electron distribution that is minimally asymmetric.
(D) Carbon is sp -hybridized in CO , making it more electronegative than a typical carbon atom.

24. Which species has the smallest tendency to dimerize?

(A) NO (B) NO_2 (C) CO_2^- (D) SO_2^-

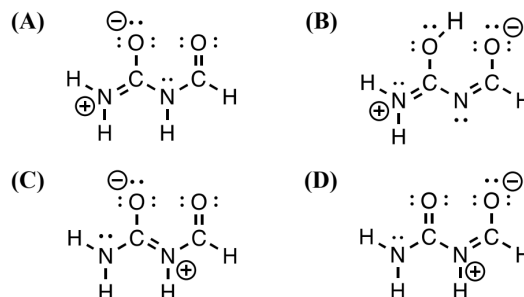
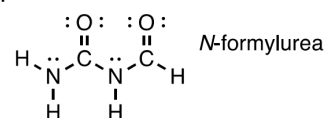
25. Which molecule has the strongest bond?

(A) H_2 (B) N_2 (C) O_2 (D) F_2

26. What is the geometry of the iodate ion, IO_3^- ?

(A) Trigonal pyramidal (B) Trigonal planar
(C) T-shaped (D) Zigzag

27. Which is NOT a valid resonance structure of *N*-formylurea?



28. Which molecule is best described as having a lone pair on nitrogen in an sp^2 -hybridized orbital?

(A) NH_3 (B) HCN
(C) H_2CNH (D) H_3CNH_2

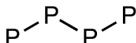
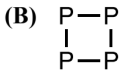
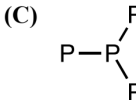
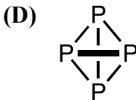
29. Nitric oxide, NO , has a smaller first ionization energy than either N atoms or O atoms. Which is the best explanation for this observation?

(A) The electron that is ionized in NO occupies an antibonding orbital.
(B) NO forms a strong bond, which lowers the ionization energy.
(C) NO has a large dipole moment, which increases electron-electron repulsion.
(D) N atoms have a half-filled $2p$ subshell, while O is very electronegative. Both of these factors give the isolated atoms unusually high ionization energies.

30. How many stereoisomers does the octahedral complex $\text{Co}(\text{NH}_3)_3(\text{CN})_3$ have?

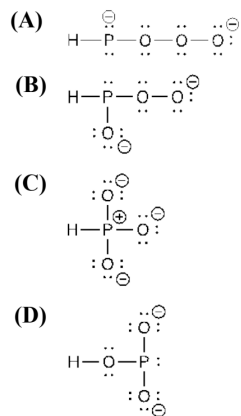
(A) 1 (B) 2 (C) 3 (D) 4

Bonding/Molecular Structure

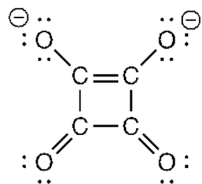
31. Which molecule has a nonzero dipole moment?
(A) CF_4 (B) C_2F_4 (C) SF_4 (D) SF_6
32. How many π bonds are in cyanogen, NCCN ?
(A) 2 (B) 4 (C) 6 (D) 7
33. In azomethane, H_3CNNCH_3 , what are the molecular geometries around the carbon and nitrogen atoms, respectively?
(A) Tetrahedral at carbon, bent at nitrogen
(B) Tetrahedral at carbon, linear at nitrogen
(C) Square planar at carbon, bent at nitrogen
(D) Square planar at carbon, linear at nitrogen
34. What is the bond order of O_2^+ ?
(A) 1.0 (B) 1.5 (C) 2.0 (D) 2.5
35. Which is the best explanation for phosphorus favoring formation of P_4 over P_2 molecules, while nitrogen forms only N_2 molecules?
(A) Phosphorus-phosphorus π bonds are weaker than nitrogen-nitrogen π bonds.
(B) Phosphorus can have an expanded octet while nitrogen cannot.
(C) The nonbonding electrons on phosphorus occupy an unhybridized s orbital while the nonbonding electrons on nitrogen occupy hybrid orbitals.
(D) The larger size of phosphorus allows the bonding electrons to be closer together.
36. In the octahedral complex $\text{Co}(\text{NH}_2\text{CH}_2\text{CO}_2)_3$, each glycinate ion $\text{NH}_2\text{CH}_2\text{CO}_2^-$ binds to cobalt through its nitrogen atom and one of its oxygen atoms. How many stereoisomers of $\text{Co}(\text{NH}_2\text{CH}_2\text{CO}_2)_3$ are possible?
(A) 2 (B) 4 (C) 6 (D) 8
37. Which species has the strongest carbon-oxygen bond?
(A) CO (B) CO_2
(C) CH_2O (D) CH_3OH
38. Which molecule has a nonzero dipole moment?
(A) O_2 (B) O_3 (C) S_8 (D) SO_3
39. Which statement best describes the structure and bonding in nitromethane, $\text{H}_3\text{C}-\text{NO}_2$?
(A) Each of the three bonds to nitrogen is a different length.
(B) The molecule is Lewis acidic because the nitrogen does not obey the octet rule.
(C) The molecule is Lewis basic because the nitrogen has a lone pair.
(D) The nitrogen atom has a trigonal planar geometry.
40. Which statements about the superoxide ion, O_2^- , are correct?
I. It has a bond order of 2.0.
II. It has exactly one unpaired electron.
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
41. Which structure best depicts the three-dimensional shape of the P_4 molecule?
(A)  (B) 
(C)  (D) 
42. How many isomers are there of the octahedral coordination complex $\text{Co}(\text{NH}_3)_3\text{Cl}_3$?
(A) 1 (B) 2 (C) 3 (D) 4
43. Which molecule is planar?
(A) CF_4 (B) COF_2 (C) SF_4 (D) SOF_2
44. Which molecule has no unpaired electrons?
(A) NO (B) O_2 (C) SO_2 (D) ClO_2

Bonding/Molecular Structure

45. Phosphorous acid, H_3PO_3 , is a diprotic acid in aqueous solution ($\text{p}K_1 = 2.0$, $\text{p}K_2 = 6.6$). Which Lewis structure best represents HPO_3^{2-} ?



46. One Lewis structure of squarate ion, $\text{C}_4\text{O}_4^{2-}$, is shown below. Which statement best describes the bond distances in the squarate ion?



- (A) All carbon-carbon bond distances are the same, and all carbon-oxygen bond distances are the same.
- (B) There are two distinct carbon-carbon bond distances, but all carbon-oxygen bond distances are the same.
- (C) There are two distinct carbon-carbon bond distances, and two distinct carbon-oxygen bond distances.
- (D) There are three distinct carbon-carbon bond distances, and two distinct carbon-oxygen bond distances.
47. The bond distance in CO is 112.8 pm, while the bond distance in NO is 115 pm. Which is the best explanation for the shorter bond in CO?
- (A) Carbon has one fewer valence electron than nitrogen.
- (B) Carbon is less electronegative than nitrogen.
- (C) Carbon has a smaller atomic radius than nitrogen.
- (D) Carbon can form up to four bonds, while nitrogen can form only three.

48. How many stereoisomers does the square planar complex $\text{Pt}(\text{NH}_3)_2(\text{Br})(\text{Cl})$ have?

(A) One (B) Two (C) Three (D) Four

49. Which statements correctly describe the geometry of the carbonate ion, CO_3^{2-} ?

I. All three carbon-oxygen bond distances are the same.
II. All three bond angles are 120° .

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

50. Which molecule has a trigonal pyramidal geometry?

(A) PCl_3 (B) BCl_3 (C) IF_3 (D) SO_3

51. The bond in F_2 has a smaller bond dissociation enthalpy than the bond in Cl_2 . Which is the best explanation for this difference?

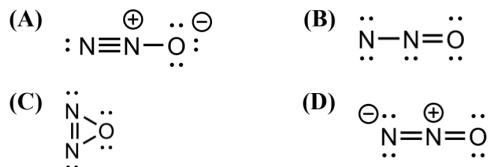
(A) Fluorine is more electronegative than chlorine.

(B) Lone pair-lone pair repulsions are stronger in F_2 than in Cl_2 .

(C) The F-F bond is longer than the Cl-Cl bond.

(D) Chlorine has energetically accessible d orbitals while fluorine does not.

52. Which resonance structure contributes the most to the overall bonding in nitrous oxide, N_2O ?



53. How many σ bonds and how many π bonds are present in allene, $\text{H}_2\text{C}=\text{C}=\text{CH}_2$?

(A) One σ , one π (B) Five σ , one π
(C) Six σ , two π (D) Seven σ , two π

54. Which aspect of metallic bonding best accounts for the characteristic malleability of metals?

(A) Metals have few valence electrons.

(B) Metals contain a lattice of positive ions.

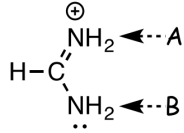
(C) Metals contain delocalized electrons.

(D) Metals have low electronegativity.

55. Which compound contains both ionic and covalent bonds?

(A) PF_3 (B) KF
(C) CH_3COOH (D) MgSO_4

Bonding/Molecular Structure

56. Which gas-phase molecule is NOT linear?
(A) CS₂ (B) SO₂ (C) HCCH (D) BrCN
57. In the Lewis structure of the chlorate ion, ClO₃⁻, how many lone pairs of electrons does the chlorine atom have?
(A) 0 (B) 1 (C) 2 (D) 3
58. A coordination complex M(NH₃)₂Cl₂ can be separated into a pair of geometric isomers. Is this observation consistent with a tetrahedral or a square planar geometry at the metal center?
(A) It is consistent only with a tetrahedral geometry.
(B) It is consistent only with a square planar geometry.
(C) It is consistent with either a square planar or a tetrahedral geometry.
(D) It is consistent with neither a square planar nor a tetrahedral geometry.
59. The bond in gas-phase O₂ (121 pm) is significantly longer than the bond in gas-phase O₂⁺ (112 pm). What is the best explanation for this difference?
(A) O₂ has one more antibonding electron than O₂⁺.
(B) O₂ has two unpaired electrons while O₂⁺ has one.
(C) The bond in O₂ has less ionic character than the bond in O₂⁺.
(D) It requires more energy to remove an electron from O₂ to form O₂⁺ than it does to remove an electron from O to form O⁺.
60. In the guanidinium ion, [C(NH₂)₃]⁺, what is the best description of the hybridizations of the nitrogen atoms?
(A) All three sp³ (B) Two sp³, one sp²
(C) One sp³, two sp² (D) All three sp²
61. Which molecule has a dipole moment of zero?
(A) CO (B) CO₂ (C) CH₂O (D) CH₃OH
62. What is the geometry of SF₄?
(A) Tetrahedral
(B) Square planar
(C) See-saw
(D) Trigonal monopyrarnidal
63. How many σ and π bonds are in 1,3-butadiene, H₂C=CH-CH=CH₂?
(A) 7 σ and 2 π bonds (B) 2 σ and 7 π bonds
(C) 9 σ and 2 π bonds (D) 2 σ and 9 π bonds
64. What is the formal charge on the central nitrogen in the Lewis structure of N₂O?
(A) 0
(B) +1
(C) -1
(D) 0 in some resonance structures, -1 in other resonance structures
65. Shown below is a Lewis structure of the formamidine ion. What geometry is exhibited by the two nitrogens in the formamidine ion?
- 
- | <u>Geometry at A</u> | <u>Geometry at B</u> |
|------------------------|----------------------|
| (A) trigonal planar | trigonal planar |
| (B) trigonal planar | trigonal pyramidal |
| (C) trigonal pyramidal | trigonal planar |
| (D) trigonal pyramidal | trigonal pyramidal |
66. How many unpaired electrons are in the superoxide ion, O₂⁻?
(A) Zero (B) One (C) Two (D) Three
67. Which molecule is nonpolar, yet contains polar covalent bonds?
(A) CO₂ (B) HCN (C) NH₃ (D) P₄
68. The nitrite ion, NO₂⁻, can be represented as a resonance hybrid of two significant Lewis structures. Which statements about this are correct?
I. The two resonance structures contribute equally to the structure.
II. The formal charge of nitrogen is zero in both resonance structures.
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
69. Which is not a stable molecule?
(A) NF₃ (B) NF₅ (C) PF₃ (D) PF₅

Bonding/Molecular Structure

70. What is the molecular geometry of IF_3 ?
 (A) Trigonal planar (B) Trigonal pyramidal
 (C) T-shaped (D) Tetrahedral
71. N_2 is a stable molecule and the N_4 molecule is unknown. P_4 is much more stable than molecular P_2 . Which is the best explanation for this difference?
 (A) N_2 has valence electrons only in bonding and nonbonding orbitals, while P_2 has some valence electrons in antibonding orbitals.
 (B) The greater electronegativity of N compared to P stabilizes compounds with lower molar masses.
 (C) The greater size of P compared to N results in decreased overlap in pi bonds.
 (D) The preference of P to adopt smaller bond angles than N favors formation of tetrahedral P_4 molecules.
72. Which contains sp^3 -hybridized carbon atoms?
 (A) Benzene, C_6H_6 (B) Ethane, C_2H_6
 (C) Ethene, C_2H_4 (D) Ethyne, C_2H_2
73. Which species are linear?
 I. NO_2^+ II. I_3^-
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
74. The peroxydisulfate anion, HSO_5^- , has
 (A) five sulfur-oxygen bonds and no oxygen-oxygen bonds.
 (B) four sulfur-oxygen bonds and one oxygen-oxygen bond.
 (C) three sulfur-oxygen bonds and two oxygen-oxygen bonds.
 (D) one sulfur-oxygen bond and four oxygen-oxygen bonds.
75. Which statement about bonding is correct?
 (A) A σ bond has cylindrical symmetry about the bonding axis.
 (B) A π bond is twice as strong as a σ bond.
 (C) A double bond consists of two π bonds.
 (D) A π bond results from the sideways overlap of hybridized orbitals.
76. What is the geometry of the chlorate ion, ClO_3^- ?
 (A) trigonal planar (B) trigonal pyramidal
 (C) T-shaped (D) zigzag
77. What is the bond order in NO?
 (A) 1.0 (B) 1.5 (C) 2.0 (D) 2.5
78. In the Lewis structure of ozone, O_3 , what is the formal charge on the central oxygen?
 (A) 2- (B) 1- (C) 0 (D) 1+
79. Which atom is least likely to violate the octet rule in its compounds?
 (A) B (B) Cl (C) F (D) H
80. In the Lewis structure for formic acid, HCOOH , how many bonding pairs and lone pairs of electrons are present?
 (A) 4 bonding, 2 lone (B) 4 bonding, 5 lone
 (C) 5 bonding, 0 lone (D) 5 bonding, 4 lone
81. Which ionic compound has the largest lattice energy?
 (A) LiF (B) BeO (C) KBr (D) CaS
82. When 1.0 mole of H_2O_2 decomposes to form H_2O and O_2 , 103 kJ of energy is released. Given the bond energies below, what is the bond energy of the O—O single bond in H_2O_2 ?
- | | | |
|--|-----|-----|
| Bond | H—O | O=O |
| Bond Energy, $\text{kJ}\cdot\text{mol}^{-1}$ | 463 | 498 |
- (A) $+395 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $+249 \text{ kJ}\cdot\text{mol}^{-1}$
 (C) $+146 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $+103 \text{ kJ}\cdot\text{mol}^{-1}$
83. Which species has a different number of pi bonds than the others?
 (A) C_2H_2 (B) CO_2 (C) N_2 (D) O_3
84. Which molecule is correctly matched with its shape as predicted by VSEPR theory?
 (A) PCl_3 trigonal pyramidal
 (B) OF_2 linear
 (C) ClF_3 trigonal planar
 (D) SF_6 hexagonal
85. The silicon-oxygen bonds in SiO_2 are best described as
 (A) coordinate covalent. (B) ionic.
 (C) nonpolar covalent. (D) polar covalent.
86. Which bond is strongest?
 (A) C=C (B) C=N (C) C=O (D) C=S

Bonding/Molecular Structure

87. What is the relationship between the two species shown below?



They are

- (A) geometric isomers. (B) enantiomers.
(C) resonance forms. (D) structural isomers.
88. If A represents the central atom, in which molecule is the F-A-F angle the smallest?
(A) BF_3 (B) CF_4 (C) NF_3 (D) OF_2

89. On the basis of VSEPR theory, what geometry is predicted for the central sulfur atom in SOCl_2 ?

- (A) bent (B) T-shaped
(C) trigonal planar (D) trigonal pyramidal

90. How many sigma (σ) and pi (π) bonds are in a molecule of ethyne (acetylene), HCCH ?

- (A) 1 σ and 1 π (B) 2 σ and 1 π
(C) 2 σ and 3 π (D) 3 σ and 2 π

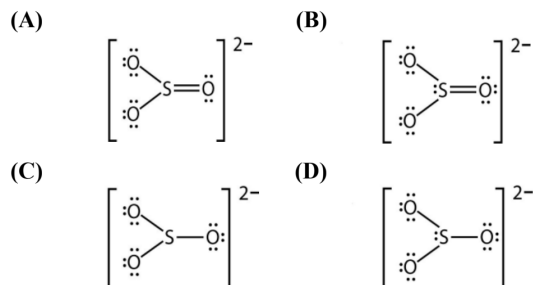
91. Which species contains only covalent bonds?

- (A) AlF_3 (B) NH_4NO_3
(C) H_2SO_4 (D) $\text{K}_2\text{Cr}_2\text{O}_7$

92. In which species can we describe the central atom as having sp^2 hybridization?

- (A) BeF_2 (B) CO_2 (C) KrF_2 (D) SO_2

93. Which Lewis dot structure is a valid representation for the sulfite ion, $[\text{SO}_3]^{2-}$?



94. A triple bond is found in which of the following species?

I CO II C_2H_2 III CN^-

- (A) I only (B) II only
(C) I and II only (D) I, II and III

95. Which of the following compounds has a non-zero dipole moment?

- (A) CO_2 (B) AsH_3 (C) CCl_4 (D) PF_5

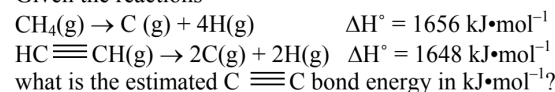
96. Which species has the largest bond angle?

- (A) NO_2^+ (B) NO_2 (C) NO_2^- (D) NO_3^-

97. Which species is planar?

- (A) CO_3^{2-} (B) SO_3^{2-} (C) ClO_3^- (D) BF_4^-

98. Given the reactions



what is the estimated $\text{C}\equiv\text{C}$ bond energy in $\text{kJ}\cdot\text{mol}^{-1}$?

- (A) 414 (B) 820 (C) 1234 (D) 1664

99. Which bonds are formed by a carbon atom with sp^2 hybridization?

- (A) 4 π bonds
(B) 2 π bonds and 2 σ bonds
(C) 1 π bond and 3 σ bonds
(D) 4 σ bonds

100. How many resonance forms can be written for the nitrate ion, NO_3^- ?

- (A) 1 (B) 2 (C) 3 (D) 4

101. How many valence electrons are in a persulfate ion, SO_5^{2-} ?

- (A) 32 (B) 34 (C) 36 (D) 38

102. Which diatomic molecule contains the strongest bond?

- (A) $\text{H}-\text{Cl}$ (B) $\text{H}-\text{F}$ (C) $\text{Cl}-\text{Cl}$ (D) $\text{F}-\text{F}$

103. The O-N-O bond angles in the nitrate ion, NO_3^- , are best described as being

- (A) all 120° (B) all 109.5°
(C) all 90° (D) two 90° , one 180°

104. Which molecule has no permanent dipole moment?

- (A) BCl_3 (B) NCl_3 (C) CHCl_3 (D) PCl_3

Bonding/Molecular Structure

105. When the compounds HF, H₂O, NH₃, and CH₄ are listed in order of increasing boiling point, which order is correct?
- (A) CH₄ < NH₃ < H₂O < HF
 (B) NH₃ < CH₄ < H₂O < HF
 (C) HF < CH₄ < H₂O < NH₃
 (D) CH₄ < NH₃ < HF < H₂O
106. Which ionic solid has the greatest lattice energy?
- (A) NaCl (B) MgO (C) KBr (D) SrS
107. In which molecule does the chlorine have the most positive partial charge?
- (A) HCl (B) BrCl (C) OCl₂ (D) SCl₂
108. Which molecule contains the shortest carbon-carbon bonds?
- (A) C₂H₂ (B) C₂H₄ (C) C₃H₈ (D) C₆H₁₂
109. How many valence electrons are in one ion of thiosulfate, S₂O₃²⁻?
- (A) 26 (B) 28 (C) 30 (D) 32
110. Which substance has the highest melting point?
- (A) CO (B) CO₂ (C) SiO₂ (D) P₂O₅
111. Which species has exactly five pairs of electrons around the central atom?
- (A) ClF₅ (B) SF₄ (C) SF₅⁻ (D) XeF₄
112. What are the hybridizations of the carbon atoms labeled C₁ and C₂, respectively, in glycine?
- C₁ C₂

(A) sp² sp²

(B) sp² sp³

(C) sp³ sp²

(D) sp³ sp³
113. What is the total number of valence electrons in the thiosulfate ion, S₂O₃²⁻?
- (A) 28 (B) 30 (C) 32 (D) 34
114. For which species is the electron pair geometry around the central atom of the Lewis dot structure the same as the geometry of the atoms?
- (A) CO₂ (B) SO₂ (C) BrO₂⁻ (D) NO₂⁻
115. Which molecule is least stable?
- (A) OF₂ (B) OF₄ (C) SF₂ (D) SF₄
116. In which choice are the molecules listed in order of increasing bond angle?
- (A) H₂O, CH₄, NH₃ (B) CH₄, NH₃, H₂O
 (C) H₂O, NH₃, CH₄ (D) NH₃, CH₄, H₂O
117. What is the shape of the ClF₃ molecule?
- (A) trigonal planar (B) trigonal pyramidal
 (C) T-shaped (D) tetrahedral
118. In which choice are the sodium halides listed in order of increasing lattice energy?
- (A) NaF, NaCl, NaBr (B) NaBr, NaCl, NaF
 (C) NaCl, NaF, NaBr (D) NaCl, NaBr, NaF
119. In which species does the central atom have one or more lone pairs of valence electrons?
- (A) AlCl₄⁻ (B) CO₂ (C) PCl₄⁺ (D) SO₂
120. Which substance has both covalent and ionic bonds?
- (A) NH₄Br(s) (B) KI(s)
 (C) CH₂Cl₂(l) (D) SiF₄(g)
121. Which has the largest bond dissociation energy?
- (A) H-F (B) H-Cl (C) H-Br (D) H-I
122. The O-N-O bond angle in the nitrite ion, NO₂⁻, is closest to
- (A) 180°. (B) 150°. (C) 120°. (D) 109°.
123. Which is a resonance form of the Lewis structure shown here?
- I.

(A) I only

II.

(B) II only
- (C) Both I and II (D) Neither I nor II

Bonding/Molecular Structure

124. What is the geometry of the fluorine atoms around the boron atom in BF_4^- ?
- (A) planar (B) see-saw
(C) tetrahedral (D) triangular pyramidal

125. Which ionic compound has the smallest lattice energy?
- (A) NaF (B) MgO (C) AlN (D) MgCl_2

126. Which species have one or more atoms that violate the octet rule?

I. NO
II. SF_2
III. PF_4^+

- (A) I only (B) III only
(C) I and II only (D) II and III only

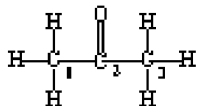
127. In which species is the carbon-nitrogen bond the shortest?

- (A) CH_3NH_2 (B) CH_2NH
(C) $(\text{CH}_3)_4\text{N}^+$ (D) CH_3CN

128. In which species is resonance most useful in explaining the observed bond lengths?

- (A) NF_3 (B) NH_4^+ (C) NO_2^+ (D) NO_2^-

129. Which description best represents the hybridization of the C_1 and C_2 atoms?



- (A) $\text{C}_1 - \text{sp}^2$, $\text{C}_2 - \text{sp}^2$
(B) $\text{C}_1 - \text{sp}^3$, $\text{C}_2 - \text{sp}^3$
(C) $\text{C}_1 - \text{sp}^3$, $\text{C}_2 - \text{sp}^2$
(D) $\text{C}_1 - \text{sp}^2$, $\text{C}_2 - \text{sp}$

130. Polar molecules include which of those listed?

I. CO_2
II. COCl_2
III. CH_2Cl_2

- (A) I only (B) I and II only
(C) I and III only (D) II and III only

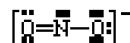
131. In which species is the central atom NOT surrounded by exactly 8 valence electrons?

- (A) BF_4^- (B) NCl_3 (C) PCl_4^+ (D) SF_4

132. In which pair of formulas are both species polar?

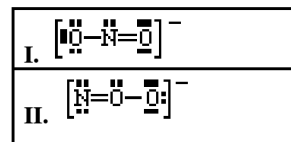
- (A) CO_2 and H_2O (B) BF_3 and PCl_3
(C) SO_2 and SCl_2 (D) CS_2 and NO_2

133. The nitrite ion can be represented;



Which of the structures represent possible resonance forms of this ion?

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II



134. Which statement is true about the most stable Lewis structure for CS_2 ?

- (A) There are no lone pairs.
(B) All bonds are double bonds.
(C) The central atom does not have an octet of electrons.
(D) A sulfur atom must be the central atom for the structure to be stable.

135. When the molecules N_2 , O_2 , F_2 are arranged in order of increasing bond strength, which order is correct?

- (A) N_2 , O_2 , F_2 (B) N_2 , F_2 , O_2
(C) O_2 , N_2 , F_2 (D) F_2 , O_2 , N_2

136. Which molecule contains three sigma (σ) and two pi (π) bonds?

- (A) C_2H_2 (B) C_2H_4 (C) C_2H_6 (D) C_3H_4

137. Which species contains only covalent bonds?

- (A) H_2SO_4 (B) NH_4NO_3
(C) NaOCl (D) K_2CrO_4

138. How many valence electrons are in the pyrophosphate ion, $\text{P}_2\text{O}_7^{4-}$?

- (A) 48 (B) 52 (C) 54 (D) 56

139. Which species has the largest F-A-F bond angle where A is the central atom?

- (A) BF_3 (B) CF_4 (C) NF_3 (D) OF_2

140. The triple bond in carbon monoxide consists of

- (A) 3 sigma bonds
(B) 2 sigma bonds and 1 pi bond
(C) 1 sigma bond and 2 pi bonds
(D) 3 pi bonds

Bonding/Molecular Structure

141. The boiling points of the halogens, F_2 , Cl_2 , Br_2 and I_2 , increase in that order. This is best attributed to differences in
- (A) covalent bond strengths.
(B) dipole forces.
(C) London dispersion forces.
(D) colligative forces.
142. Which species is polar?
- (A) CO_2 (B) SO_2 (C) SO_3 (D) O_2
143. All of the following lists include at least one ionic compound EXCEPT
- (A) NO_2 , $NaNO_2$, KNO_3 (B) CF_4 , CaF_2 , HF
(C) $NaCl$, $MgCl_2$, SCl_2 (D) H_2S , SO_2 , SF_6
144. Which species below has the same general shape as NH_3 ?
- (A) SO_3^{2-} (B) CO_3^{2-} (C) NO_3^- (D) SO_3
145. When forming covalent bonds, which atom can have more than eight valence electrons?
- (A) H (B) N (C) F (D) Cl
146. Which diatomic molecule has the shortest bond length?
- (A) N_2 (B) O_2 (C) F_2 (D) S_2
147. Which species is nonpolar?
- (A) HCl (B) OCI_2 (C) NCl_3 (D) CCl_4
148. In which species are all the carbon atoms considered to be sp^2 hybridized?
- (A) C_2H_2 (B) C_2H_4 (C) C_3H_8 (D) C_4H_{10}
149. Which Lewis dot structure is the best representation of the bonding in the thiocyanate ion, SCN^- ?
- (A) $[\text{:}\ddot{\text{S}}\text{:C}::\ddot{\text{N}}\text{:}]^-$ (B) $[\text{:}\ddot{\text{S}}\text{:}\ddot{\text{C}}::\ddot{\text{N}}\text{:}]^-$
(C) $[\text{:}\ddot{\text{S}}::\ddot{\text{C}}::\ddot{\text{N}}\text{:}]^-$ (D) $[\text{:}\ddot{\text{S}}::\text{C}::\ddot{\text{N}}\text{:}]^-$
150. Which pair of atoms forms the most ionic bond?
- (A) Al and As (B) Al and N
(C) Al and Se (D) Al and O
151. In which species does the central atom obey the octet rule?
- (A) XeF_4 (B) SF_4
(C) SiF_4 (D) ClF_4^-
152. According to VSEPR theory, in which species do all the atoms lie in the same plane?
1. CH_3^+ 2. CH_3^-
- (A) 1 only (B) 2 only
(C) both 1 and 2 (D) neither 1 nor 2
153. The $H-O-H$ bond angles in H_3O^+ are approximately 107° . The orbitals used by oxygen in these bonds are best described as
- (A) p orbitals. (B) sp hybrid orbitals.
(C) sp^2 hybrid orbitals. (D) sp^3 hybrid orbitals.
154. Which bond is expected to be the *least* polar?
- (A) $O-F$ (B) $P-F$
(C) $Si-N$ (D) $B-Cl$
155. For which element would XH_3 be a stable species?
- (A) C (B) Cl (C) P (D) S
156. Which species has a Lewis diagram and structure most like that of the carbonate ion, CO_3^{2-} ?
- (A) NO_3^- (B) CH_3^+
(C) SO_3^{2-} (D) PO_4^{3-}
157. Which compounds contain both ionic and covalent bonds?
- I. $BaSO_4$ II. $Ca(NO_3)_2$ III. NH_4Cl
- (A) I only (B) I and III only
(C) II and III only (D) I, II and III
158. When the species NH_4^+ , H_3O^+ , and H_2F^+ are arranged in order of increasing $H-X-H$ bond angles, what is the correct order?
- (A) $H_3O^+ < NH_4^+ < H_2F^+$ (B) $H_2F^+ < H_3O^+ < NH_4^+$
(C) $NH_4^+ < H_2F^+ < H_3O^+$ (D) $NH_4^+ < H_3O^+ < H_2F^+$

Bonding/Molecular Structure

159. The nitrite ion, NO_2^- , may be represented by two major resonance forms. The lengths of the nitrogen-to-oxygen bonds in this ion are expected to be
- (A) the same as the length of nitrogen-to-oxygen double bonds.
 - (B) the same as the length of nitrogen-to-oxygen triple bonds.
 - (C) between the lengths of a nitrogen-to-oxygen single bond and a nitrogen-to-oxygen double bond.
 - (D) between the lengths of a nitrogen-to-oxygen double bond and a nitrogen-to-oxygen triple bond.
160. Which set contains only covalently bonded molecules?
- (A) BCl_3 , SiCl_4 , PCl_3
 - (B) NH_4Br , N_2H_4 , HBr
 - (C) I_2 , H_2S , NaI
 - (D) Al , O_3 , As_4
161. What is the total number of valence electrons in the chlorate ion, ClO_3^- ?
- (A) 24
 - (B) 26
 - (C) 28
 - (D) 32
162. The concept of resonance is used to describe molecular structures which
- (A) oscillate between two structures.
 - (B) have mirror images.
 - (C) can be isolated in several isomeric forms.
 - (D) have more than one possible Lewis structure.
163. What is the arrangement in space of the hybrid orbitals of an atom with sp^2 hybridization?
- (A) linear
 - (B) bent
 - (C) pyramidal
 - (D) trigonal planar
164. Which species is isoelectronic with NO_2^+ ?
- (A) N_2O
 - (B) NO_2^-
 - (C) NH_2^-
 - (D) SO_2
165. Which species can form intermolecular hydrogen bonds with other molecules or ions of the same type?
1. HF 2. CH_3F 3. NH_4^+
- (A) 1 only
 - (B) 3 only
 - (C) 1 and 3 only
 - (D) 1, 2 and 3